

Algorithms
gate 2025
crk, DA

$O(1)$
 $O(\text{indep})$

Asymptotic
notch.
 $\Theta, \Omega, O, o, \omega$

Lec_10

Practice Questions on time Complexity

29

29.)

Main()

```
{ int j,n;
```

```
  j=1;
```

```
  While(j<=n)
```

```
  { j=j*2;
```

```
  }
```

```
}
```

$\log n$

7 times
 $n=32$

Find the number of comparisons

Made in the execution of the loop for any

$N > 0$;

1.) n 2.) $\text{ceil}(\log n)$

3.) $\text{ceil}(\log n) + 1$

4.) $\text{floor}(\log n) + 1$

$\lceil \log n \rceil + 1$
 $\lfloor \log n \rfloor + 1$

$\log_2 16 = 4$

$\lceil \log n \rceil$
 $j <= n$

$n = 32$

$j = 1$

$j = 2$

$j = 4$

$j = 8$

$j = 16$

$j = 32$

$j = 64$

$j <= n$

$j <= n$

$j <= n$

$j <= n$

$j <= n$

$j <= n$

$j <= n$

$j = 1 \times 2$

$j = 2 \times 2$

$j = 2 \times 4$

$j = 2 \times 8$

$j = 2 \times 16$

$j = 2 \times 32$

$\text{ceil}(2.5) = 3$
 $\text{floor}(2.5) = 2$

$\log_2 32 = 5$

$\lceil \log_2 25 \rceil = 5 + 1 = 6$

~~Teil 1~~
 $\log_2 25 =$
 4.6

$n=25$

$j=1,$

$j \angle = 25,$

$j = j * 2$

$j=1,$

$1 \angle = 25, j = 1 * 2$

$j=2,$

$2 \angle = 25,$

$j = 2 * 2$

$j=4,$

$4 \angle = 25,$

$j = 8$

$j=8,$

$8 \angle = 25,$

$j = 16$

$j=16,$

$16 \angle = 25,$

$j = 32$

$j=32,$

$32 \angle = 25$

X

30.) Main()

{

for(i=1; i<=n; i++)

{ for(j=1; j<=n; j=j+i)

{

x=y+z;

}

}

}

$i=1$

for(j=1; j<=n; j=j+1) $\Rightarrow$ n times

$i=2$

for(j=1; j<=n; j=j+2) $\Rightarrow$ $n/2$ times

j=1, j=3, j=5, j=7

$i=3$

for(j=1; j<=n; j=j+3) $\Rightarrow$ $n/3$ times

1, 4, 7, 10

$$n + \frac{n}{2} + \frac{n}{3} + \frac{n}{4} + \dots + \frac{n}{n}$$

$$\Rightarrow n \left[1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n} \right]$$

$$= n \log_2 n$$

$$= \Theta(n \log n)$$

$$O(n \log n)$$

log n

31.)

For($i=n$, $j=0$; $i>0$; $i=i/2$ $j=j+1$)

Let $\text{val}(j)$ denote the value stored in the variable j after termination of the loop than find $\text{val}(j)$.

$J=0$; compare $\Rightarrow i > 0$
 for($i=n$; $i > 0$; $i=i/2$)
 $\{$ $j=j+1$
 $\}$

$$J = 0 + n$$

$$J = n + \frac{n}{2}$$

$$J = n + \frac{n}{2} + \frac{n}{4}$$

$$\frac{n}{2^n} = 1$$

$$n = 2^n$$

$J=0$ $J=J+1$

$i=n, i > 0, i=i/2$
 $i=n, n > 0, i=n/2$
 $i=n/2, n/2 > 0, i=n/4$
 $i=n/4, n/4 > 0, i=n/8$
 $\vdots$

~~$r \geq 2$~~
constant

$$T = n + \frac{n}{2} + \frac{n}{4} + \dots + \frac{n}{2^{n-1}}$$

$$= n \left[1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{n-1}} \right]$$

$$= n \left[\frac{1 \left(\frac{1}{2} \right)^n - 1}{1 - \frac{1}{2}} \right]$$

$$\frac{a(r^n - 1)}{r - 1}$$

Sum of GP

$$\frac{1}{2^{100}}$$

$= n \times \text{constant}$

$$= n$$

$O(n)$

$$r = \bar{s}/q$$

$$\textcircled{O(1)}$$

$$\left(\frac{\bar{s}}{q}\right) + \left(\frac{\bar{s}}{q}\right)^2 + \dots + \left(\frac{\bar{s}}{q}\right)^n$$

(constant)

$$O(1) \Rightarrow \textcircled{O(1)}$$

.



Thanks